

Network Analysis:

The Hidden Structures behind the Webs We Weave

17-338 / 17-668

Power and Centrality in Social Networks

Tuesday, September 23, 2025

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2-min Quiz, on Canvas



“Central” actors are powerful

Relational view of the individual

So far, we covered dyads and triads as basic building blocks of networks

The relational view has been the overarching theme

- relations between two people
- relations among three people

The same relational view to study the individual

This approach sharply contrasts with widely used approaches

- Surveys ask individuals about their attributes, beliefs, and actions
- Majority of statistical models assume that individuals are like disconnected atoms
 - Independent, identically distributed variables assumption (IID).

Which node is the most “prominent”?

In contrast, network analysis views an individual in relation to their network neighbors

- **Opinions** are influenced by network neighbors
- Individual's **status** and importance comes from their position in the network
- Individuals can leverage their position to **influence opinions, selectively spread or block information, and control opportunities**
- **Power** that an individual can wield is fundamentally relational

A network position of power and prominence is a “central” position

- Power and prominence are multifaceted
- Therefore, “central” can reflect different aspects of power
 - “Centrality” can be defined and measured in different ways

So many ways to define “centrality”

Centrality

Degree

<code>degree centrality</code> (G)	Compute the degree centrality for nodes.
<code>in_degree centrality</code> (G)	Compute the in-degree centrality for nodes.
<code>out_degree centrality</code> (G)	Compute the out-degree centrality for nodes.

Eigenvector

<code>eigenvector centrality</code> (G[, max_iter, tol, ...])	Compute the eigenvector centrality for the graph G.
<code>eigenvector centrality_numpy</code> (G[, weight, ...])	Compute the eigenvector centrality for the graph G.
<code>katz centrality</code> (G[, alpha, beta, max_iter, ...])	Compute the Katz centrality for the nodes of the graph G.
<code>katz centrality_numpy</code> (G[, alpha, beta, ...])	Compute the Katz centrality for the graph G.

Closeness

<code>closeness centrality</code> (G[, u, distance, ...])	Compute closeness centrality for nodes.
<code>incremental_closeness centrality</code> (G, edge[, ...])	Incremental closeness centrality for nodes.

Current Flow Closeness

<code>current_flow_closeness centrality</code> (G[, ...])	Compute current-flow closeness centrality for nodes.
<code>information centrality</code> (G[, weight, dtype, ...])	Compute current-flow closeness centrality for nodes.

(Shortest Path) Betweenness

<code>betweenness centrality</code> (G[, k, normalized, ...])	Compute the shortest-path betweenness centrality for nodes.
<code>betweenness centrality_subset</code> (G, sources, ...)	Compute betweenness centrality for a subset of nodes.
<code>edge_betweenness centrality</code> (G[, k, ...])	Compute betweenness centrality for edges.
<code>edge_betweenness centrality_subset</code> (G, ...[, ...])	Compute betweenness centrality for edges for a subset of nodes.

Current Flow Betweenness

<code>current_flow_betweenness centrality</code> (G[, ...])	Compute current-flow betweenness centrality for nodes.
<code>edge_current_flow_betweenness centrality</code> (G)	Compute current-flow betweenness centrality for edges.
<code>approximate_current_flow_betweenness centrality</code> (G)	Compute the approximate current-flow betweenness centrality for nodes.
<code>current_flow_betweenness centrality_subset</code> (G, ...)	Compute current-flow betweenness centrality for subsets of nodes.
<code>edge_current_flow_betweenness centrality_subset</code> (G, ...)	Compute current-flow betweenness centrality for edges using subsets of nodes.

Communicability Betweenness

<code>communicability_betweenness centrality</code> (G)	Returns subgraph communicability for all pairs of nodes in G.
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Group Centrality

<code>group_betweenness centrality</code> (G, C[, ...])	Compute the group betweenness centrality for a group of nodes.
<code>group_closeness centrality</code> (G, S[, weight])	Compute the group closeness centrality for a group of nodes.
<code>group_degree centrality</code> (G, S)	Compute the group degree centrality for a group of nodes.
<code>group_in_degree centrality</code> (G, S)	Compute the group in-degree centrality for a group of nodes.
<code>group_out_degree centrality</code> (G, S)	Compute the group out-degree centrality for a group of nodes.
<code>prominent_group</code> (G, k[, weight, C, ...])	Find the prominent group of size k in graph G .

Load

<code>load centrality</code> (G[, v, cutoff, normalized, ...])	Compute load centrality for nodes.
<code>edge_load centrality</code> (G[, cutoff])	Compute edge load.

Subgraph

<code>subgraph centrality</code> (G)	Returns subgraph centrality for each node in G.
<code>subgraph centrality_exp</code> (G)	Returns the subgraph centrality for each node of G.
<code>estrada_index</code> (G)	Returns the Estrada index of a the graph G.

Harmonic Centrality

<code>harmonic centrality</code> (G[, nbunch, distance, ...])	Compute harmonic centrality for nodes.
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Dispersion

<code>dispersion</code> (G[, u, v, normalized, alpha, b, c])	Calculate dispersion between u and v in G .
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Reaching

<code>local_reaching centrality</code> (G, v[, paths, ...])	Returns the local reaching centrality of a node in a directed graph.
<code>global_reaching centrality</code> (G[, weight, ...])	Returns the global reaching centrality of a directed graph.

Percolation

<code>percolation centrality</code> (G[, attribute, ...])	Compute the percolation centrality for nodes.
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Second Order Centrality

<code>second_order centrality</code> (G)	Compute the second order centrality for nodes of G.
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Trophic

<code>trophic_levels</code> (G[, weight])	
<code>trophic_differences</code> (G[, weight])	
<code>trophic_incoherence_parameter</code> (G[, weight])	

VoteRank

<code>voterank</code> (G[, number_of_nodes])	Select
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Laplacian

<code>laplacian centrality</code> (G[, normalized, ...])	
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...

So many ways to define “centrality”

We will cover the four most widely used centrality measures in network analysis

- Degree centrality
- Closeness centrality
- Betweenness centrality
- Eigenvector centrality

Degree Centrality

Getting Past the Gatekeeper:



Insight: **Central nodes have many connections**

- Access to information
 - Gate keepers

Degree Centrality

Insight: **Central nodes have many connections**

- Access to information
 - Gate keepers
- Concentration of attention (attention is scarce)
 - Opinion leaders



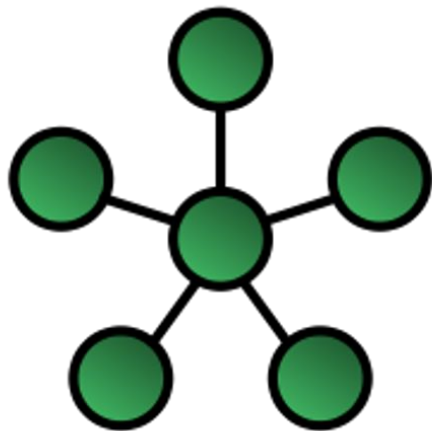
Degree Centrality

Insight: **Central nodes have many connections**

- Access to information
 - Gate keepers
- Concentration of attention (attention is scarce)
 - Opinion leaders
- Signal of “quality” under high uncertainty
 - Influencers
 - ‘There must be some reason why he has so many connections.’



Degree Centrality



Insight: Central nodes have many connections

The number of ties of a node, **normalized by network size** ($g-1$).

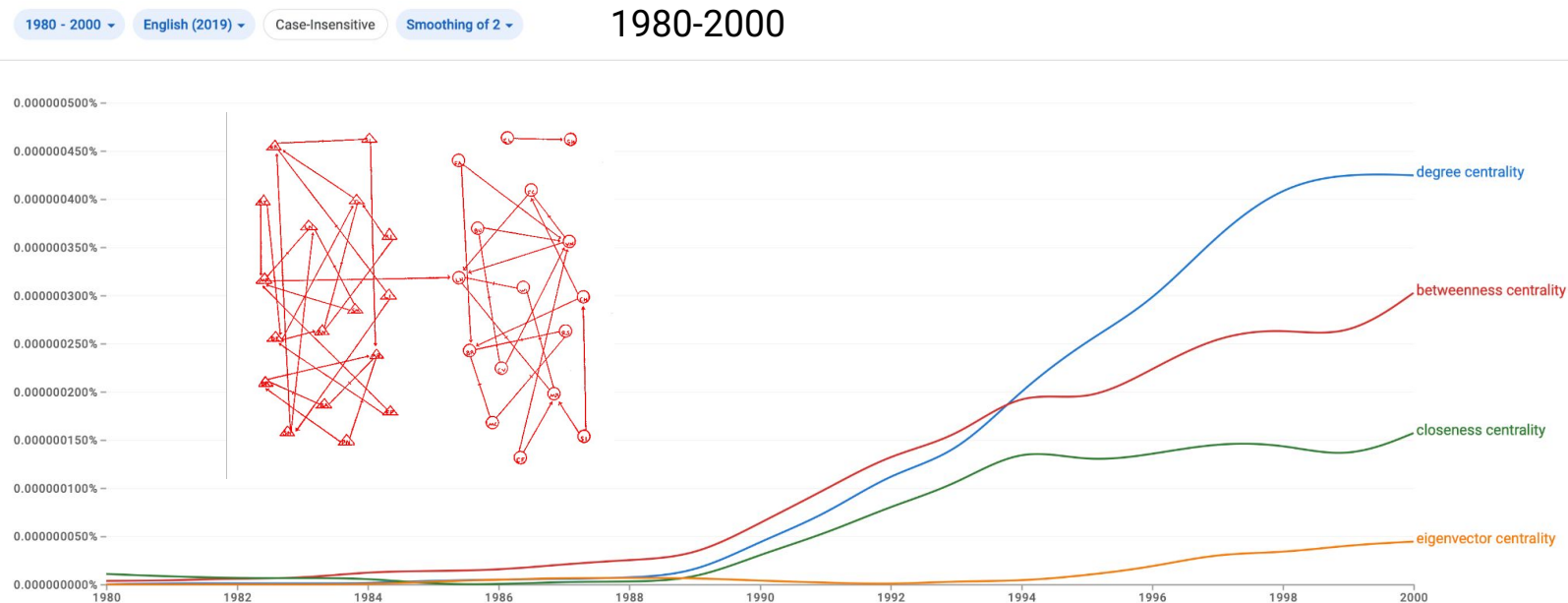
$$C_D(i) = \frac{k_i}{g - 1}$$

Normalization allows for comparison across different networks

One of the most basic and fundamental quantities in network science

Degree Centrality

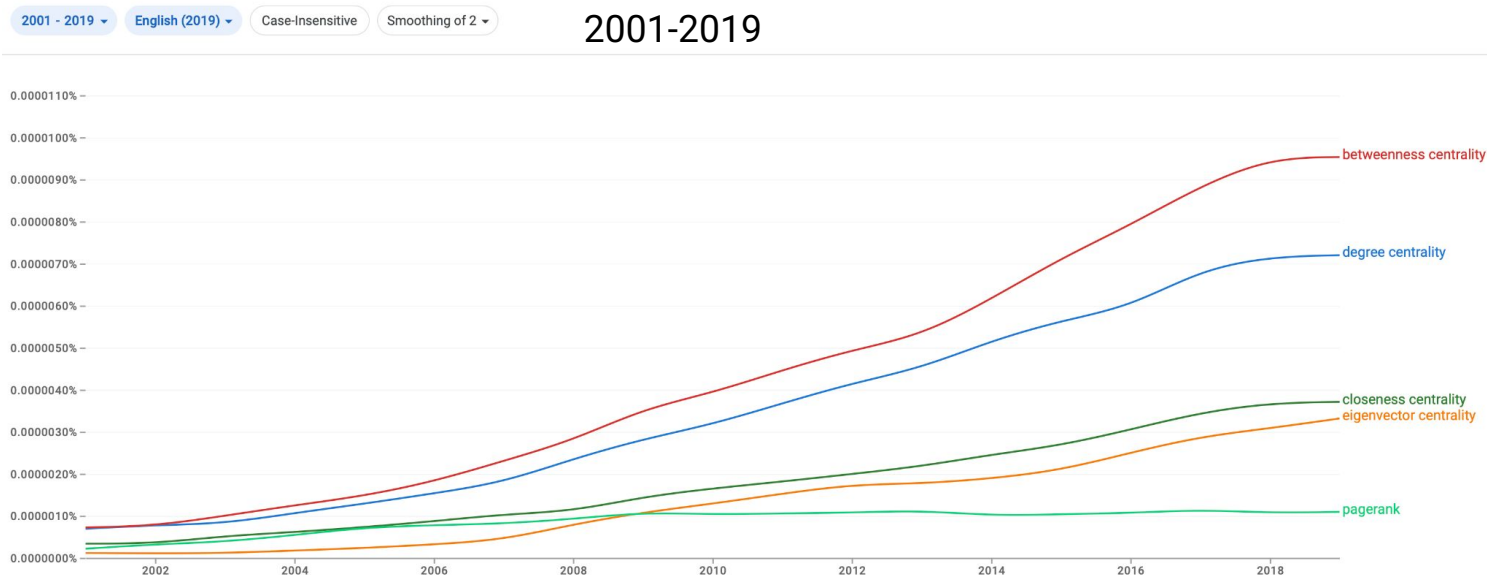
Degree centrality was, by far, the most popular measure up to the early 2000s



Degree Centrality

However, its prominence arguably diminished in the past two decades

Why?

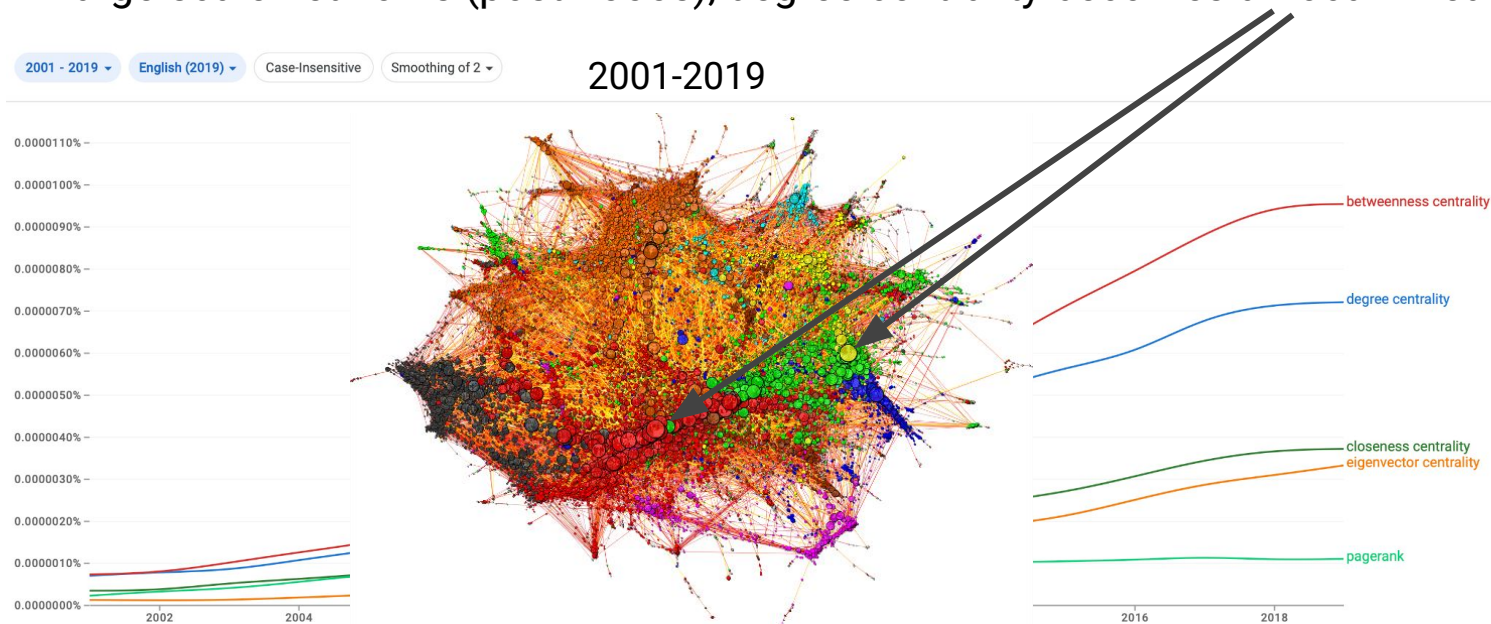


Degree Centrality

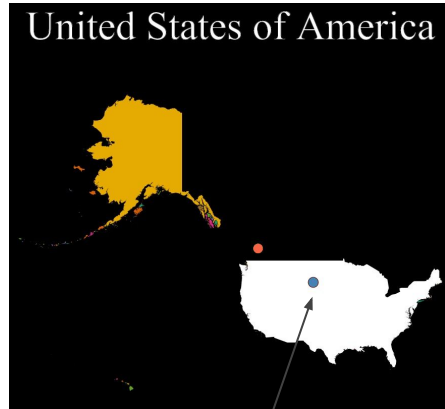
However, its prominence arguably diminished in the past two decades

Why?

In large-scale networks (post-2000s), degree centrality becomes a “local” measure



Closeness Centrality

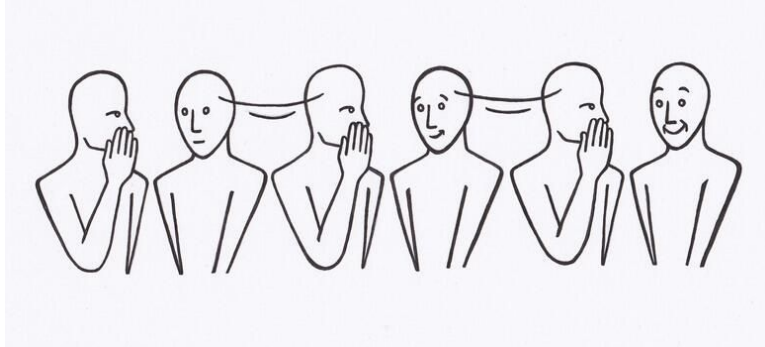


Centroid of the U.S.

Insight: The position that minimizes the distance to all other positions is the most central

Closeness Centrality

Insight: The position that minimizes the distance to all other positions is the most central



Telephone game

Application:

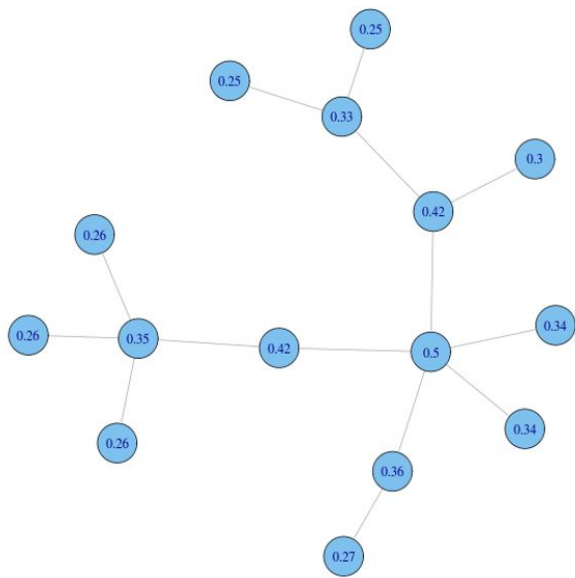
- Systems where traversing the network is costly
- Information attrition rate is high

→ The node closest to all other nodes can obtain more accurate messages

→ Faster access

→ A source of power

Closeness Centrality



Central nodes have short paths to other nodes

Calculated as the reciprocal of the sum of pairwise path distances

$$C_c(i) = \left[\sum_{j=1}^g d(i, j) \right]^{-1}$$

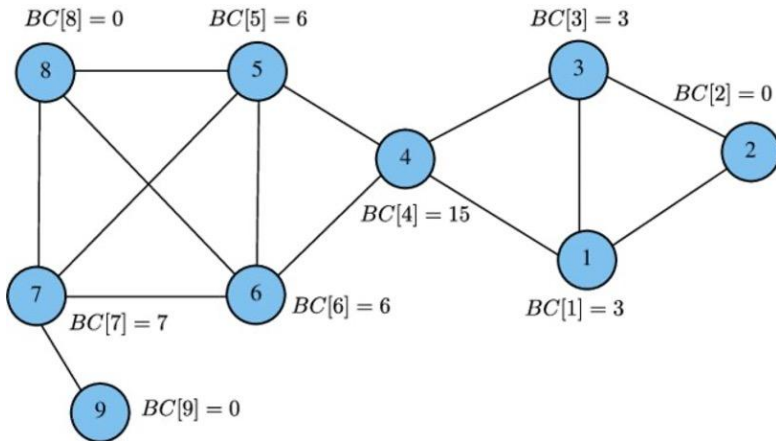
Normalized by size of network

$$C_c(i) = (g - 1) \left[\sum_{j=1}^g d(i, j) \right]^{-1}$$

A global measure that uses information from the entire network

Betweenness Centrality

Number of shortest paths node is on



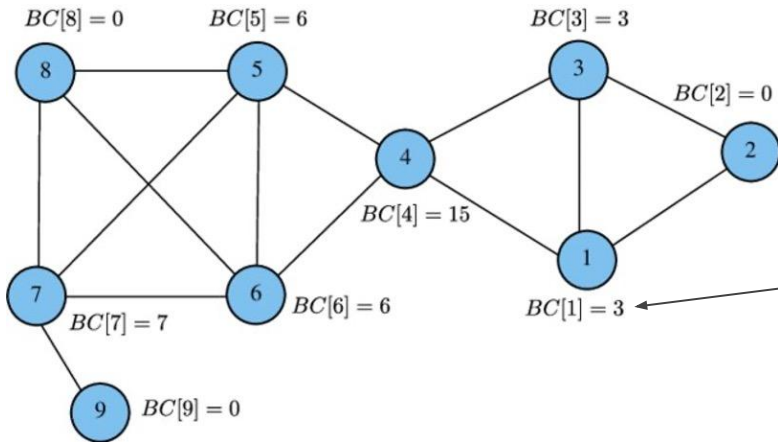
Node is central if it sits on many shortest paths

Nodes positioned at Information bottlenecks are central

Those nodes have more control over the distribution of information and other resources

Betweenness Centrality

Number of shortest paths node is on



Node i 's betweenness is the sum of the probabilities that i is on the shortest paths of all node pairs in the network

$$C_B(i) = \sum_{j < k} PL(i, j, k) / PL(j, k)$$

$PL(i, j, k) \rightarrow$ Number of shortest paths involving i

$PL(j, k) \rightarrow$ Number of shortest paths between j and k

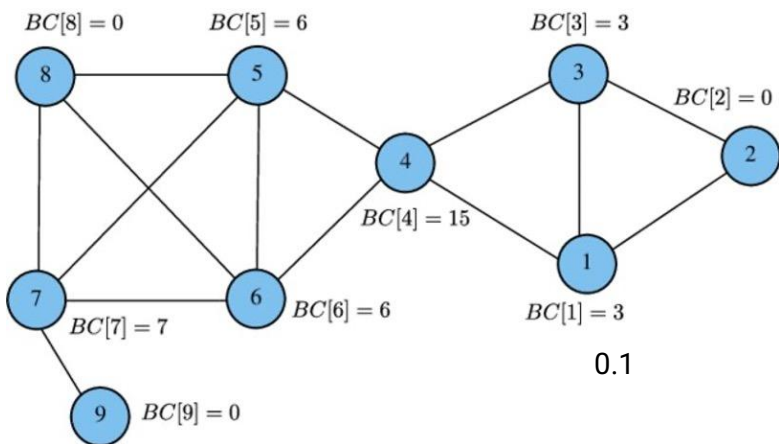
Betweenness Centrality

$$C_B(i) = \sum_{j < k} PL(i, j, k) / PL(j, k)$$

Maximum probability $PL(i, j, k) / PL(j, k) = 1$ is when i sits on every shortest path between j and k .

Then, the theoretical maximum $C(i)$ is when i sits on every shortest path for every pair of nodes (excluding i)

Q: What is the number of node pairs (dyads) excluding i in a graph with g nodes?



Betweenness Centrality

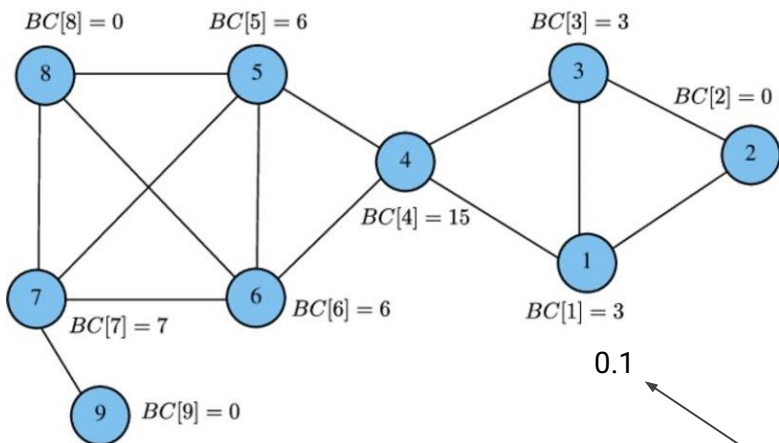
$$C_B(i) = \sum_{j < k} PL(i, j, k) / PL(j, k)$$

Maximum probability $PL(i, j, k) / PL(j, k) = 1$ is when i sits on every shortest path between j and k .

Then, the theoretical maximum $C(i)$ is when i sits on every shortest path for every pair of nodes (excluding i)

→ Normalize by the total number of node pairs excluding $i \rightarrow (g-1)(g-2)/2$

$$C_B'(i) = \frac{C_B(i)}{\left[\frac{(g-1)(g-2)}{2} \right]}$$



0.1

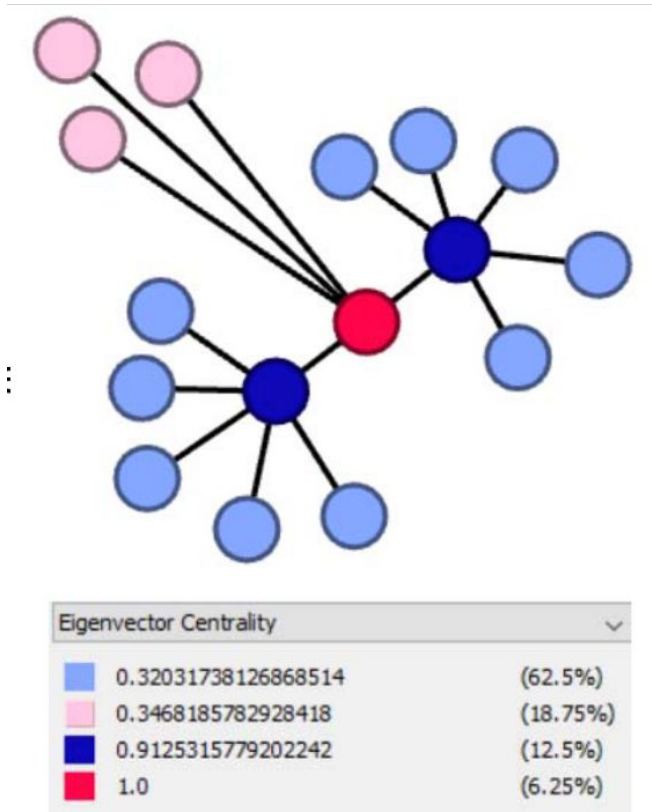
Eigenvector Centrality

Who is more central?

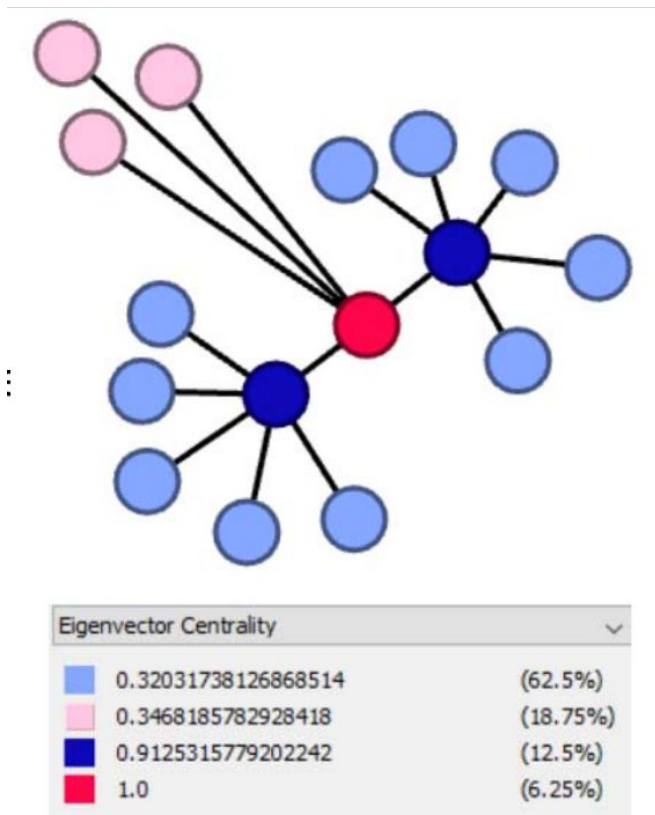
→ Someone who knows five (well connected) celebrities vs. someone who knows five ordinary people

Eigenvector centrality quantifies this insight

→ One's Eigenvector centrality is determined by the neighbors' eigenvector centrality, which in turn are determined by their neighbors' eigenvector centrality ...



Eigenvector Centrality



Who is more central?

→ Someone who knows five (well connected) celebrities vs. someone who knows five ordinary people

Eigenvector centrality quantifies this intuition

→ One's Eigenvector centrality is determined by the neighbors' eigenvector centrality, which in turn are determined by their neighbors' eigenvector centrality ...

Red node has two (purple) friends who are themselves highly connected

→ Highest Eigenvector centrality

Pink nodes have same degree centrality as the sky blue nodes, but they are connected to the red node (highest Eigenvector centrality)

→ Eigenvector centrality: pink > sky blue

Eigenvector Centrality

Given an adjacency matrix $\mathbf{A}$, eigenvector centrality of node i is:

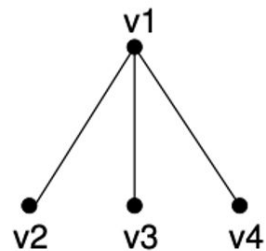
- i 's weighted degree where each of i 's edge is weighted by the degree of neighbor, j

$$\lambda C_E(i) = \sum_j A_{ij} C_E(j)$$

Given adjacency matrix $\mathbf{A}$, compute the eigenvector $\mathbf{x}$ corresponding to the principal eigenvalue λ^* of $\mathbf{A}$

Each element in $\mathbf{x}$ is the eigenvector centrality value of the corresponding node

Eigenvector Centrality



$$\begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$

For adjacency matrix A:

Eigenvector centrality of the nodes are

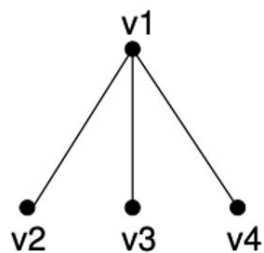
$$x_1 = \frac{1}{\lambda}(x_2 + x_3 + x_4)$$

$$x_2 = \frac{1}{\lambda}(x_1)$$

$$x_3 = \frac{1}{\lambda}(x_1)$$

$$x_4 = \frac{1}{\lambda}(x_1)$$

Eigenvector Centrality



$$\begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$

$$\lambda x_1 = (x_2 + x_3 + x_4)$$

$$\lambda x_2 = x_1$$

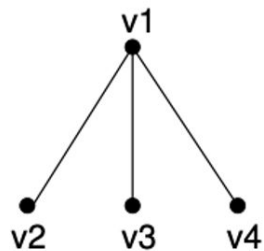
$$\lambda x_3 = x_1$$

$$\lambda x_4 = x_1$$



$$\lambda \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$$

Eigenvector Centrality



$$\begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$

$$\lambda x_1 = (x_2 + x_3 + x_4)$$

$$\lambda x_2 = x_1$$

$$\lambda x_3 = x_1$$

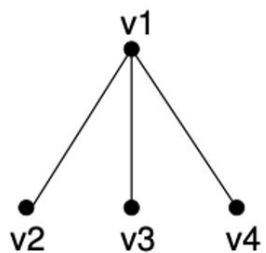
$$\lambda x_4 = x_1$$



$$\lambda \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$$


$$\lambda x = Ax$$

Eigenvector Centrality



$$\begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$

$$\lambda \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} \quad \longrightarrow \quad \lambda \mathbf{x} = \mathbf{A} \mathbf{x}$$

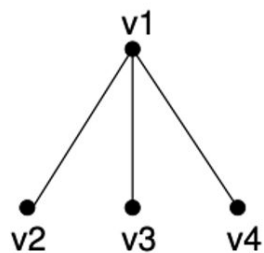
$$\mathbf{A} \mathbf{x} - \lambda \mathbf{x} = (\mathbf{A} - \lambda \mathbf{I}) \mathbf{x} = 0$$

$$\mathbf{A} - \lambda \mathbf{I} = \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} - \begin{bmatrix} \lambda & 0 & 0 & 0 \\ 0 & \lambda & 0 & 0 \\ 0 & 0 & \lambda & 0 \\ 0 & 0 & 0 & \lambda \end{bmatrix} = \begin{bmatrix} -\lambda & 1 & 1 & 1 \\ 1 & -\lambda & 0 & 0 \\ 1 & 0 & -\lambda & 0 \\ 1 & 0 & 0 & -\lambda \end{bmatrix}$$

For matrix $\mathbf{A}$, characteristic equation in λ is:

$$\begin{bmatrix} -\lambda & 1 & 1 & 1 \\ 1 & -\lambda & 0 & 0 \\ 1 & 0 & -\lambda & 0 \\ 1 & 0 & 0 & -\lambda \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

Eigenvector Centrality



$$\begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$

$$\lambda \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} \quad \longrightarrow \quad \lambda x = Ax$$

Solution to the characteristic equation:

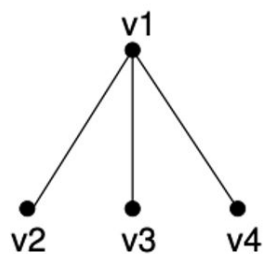
→ determinant of $A - \lambda I$ is 0

$$\lambda^4 - 3\lambda^2 = \lambda^2(\lambda^2 - 3) = 0$$

Eigenvalues are $-\sqrt{3}, 0, 0, \sqrt{3}$

Principal eigenvalue $\lambda^* = \sqrt{3}$

Eigenvector Centrality



$$\begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$

$$\lambda x_1 = (x_2 + x_3 + x_4)$$

$$\lambda x_2 = x_1$$

$$\lambda x_3 = x_1$$

$$\lambda x_4 = x_1$$



$$\sqrt{3}x_1 = x_2 + x_3 + x_4$$

$$\sqrt{3}x_2 = x_1$$

$$\sqrt{3}x_3 = x_1$$

$$\sqrt{3}x_4 = x_1$$

$$\mathbf{x} = \begin{bmatrix} \sqrt{3} \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

v1 has highest eigenvector centrality, $\sqrt{3}$

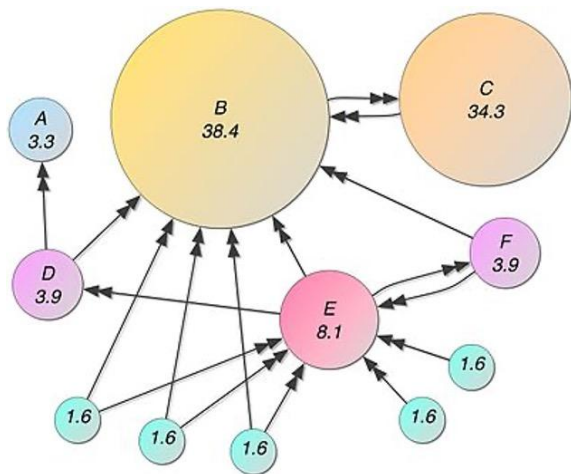
while $v_2 = v_3 = v_4 = 1$

$$\lambda \mathbf{x} = \mathbf{A}\mathbf{x} \longrightarrow \sqrt{3} \begin{bmatrix} \sqrt{3} \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \sqrt{3} \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

Eigenvector Centrality

Google's original PageRank algorithm ranks web pages using a variant of eigenvector centrality

- A node's PageRank is a function of the PageRank of its in-neighbors
- Normalize the eigenvector centrality of the in-neighbor nodes with their outdegree



$$R(w) = \lambda \sum_{v \in C_w} \frac{R(v)}{L_v}$$

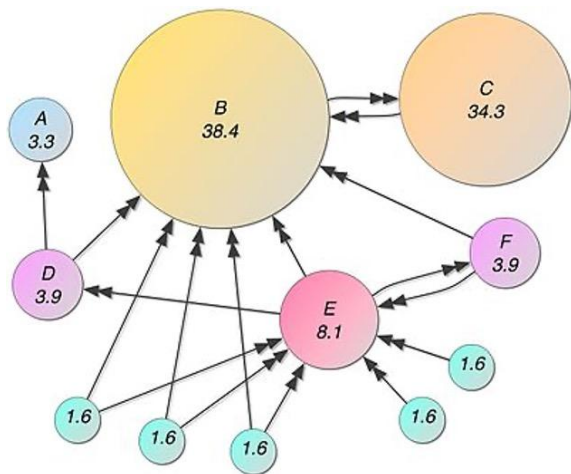
Annotations for the equation:

- $R(w)$: PageRank of w
- λ : lambda
- $\sum_{v \in C_w}$: Node w 's in-link neighbors
- L_v : Node v 's outdegree

Eigenvector Centrality

Google's original PageRank algorithm ranks web pages using a variant of eigenvector centrality

- A node's PageRank is a function of the PageRank of its in-link neighbors
- **Normalize the eigenvector centrality** of the in-link neighbors with their outdegrees
- **Why?**



$$R(w) = \lambda \sum_{v \in C_w} \frac{R(v)}{L_v}$$

Annotations for the equation:

- $R(w)$: PageRank of w
- λ : lambda
- $\sum_{v \in C_w}$: Node w 's in-link neighbors
- L_v : Node v 's outdegree

Similarities among Centrality Measures

Average correlations between centrality measures (N=58).

Pearson correlation

		1	2	3	4	5	6	7	8	9	10	11	
1	Indegree												
2	Outdegree	0.3											
3	Degree	0.78	0.71										
4	Between	0.62	0.54	0.7									
5	S-Between	0.69	0.5	0.85	0.67								
6	Closeness-In	0.55	0.16	0.45	0.37	0.31							
7	Closeness-Out	0.18	0.81	0.56	0.39	0.38	0.02						
8	S-Closeness	0.4	0.64	0.66	0.37	0.44	0.42	0.65					
9	Integration	0.7	0.26	0.58	0.5	0.41	0.9	0.15	0.51				
10	Radiality	0.21	0.86	0.61	0.44	0.41	0.06	0.98	0.67	0.19			
11	S-Int/Rad	0.45	0.7	0.73	0.43	0.5	0.44	0.69	0.99	0.54	0.72		
12	Eigenvector	0.71	0.69	0.92	0.64	0.72	0.44	0.55	0.63	0.57	0.59	0.71	
Average		0.51	0.56	0.69	0.52	0.53	0.37	0.49	0.58	0.48	0.52	0.63	0.65
Standard Deviation		0.21	0.23	0.14	0.16	0.14	0.27	0.22	0.25	0.28	0.17	0.12	0.12

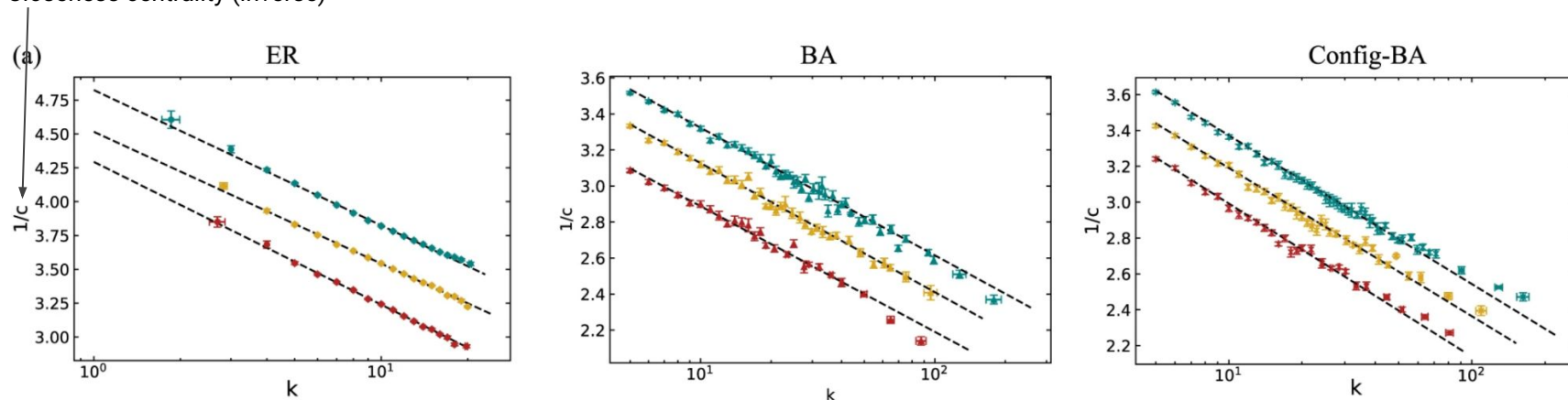
**** S-Between:**
Betweenness centrality
on symmetrized network

Similarities among Centrality Measures

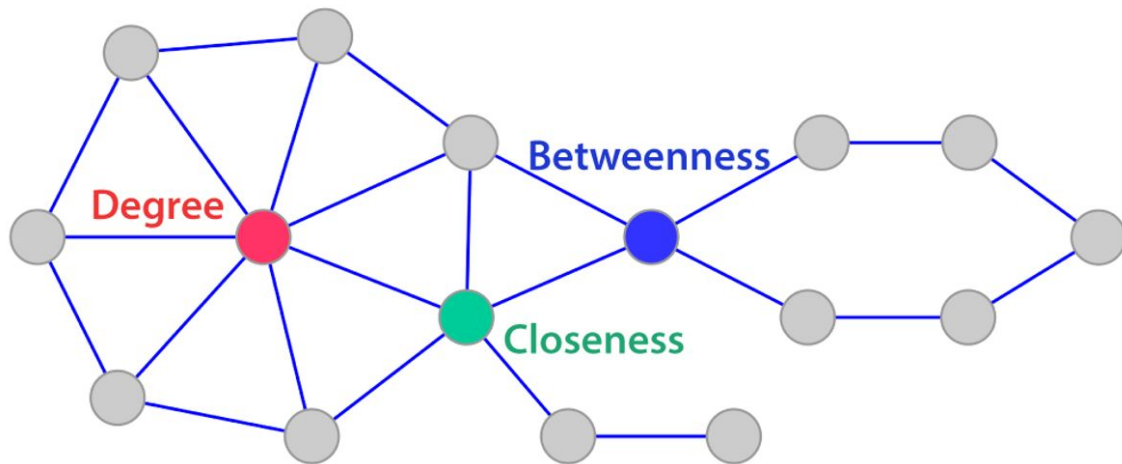
Sometimes, the relationship is non-linear $\rightarrow$ lower linear correlation (Pearson)

- Example: Closeness centrality and the log of degree centrality are highly correlated (redundant)
- If such a strong relationship, why do we care about computationally expensive closeness centrality?

Closeness centrality (inverse)



Similar, but also different



Different centrality measures capture different intuitions

In a graph, the node with highest degree is not necessarily the node with the highest betweenness or closeness

Degree	ClosenessCentrality	BetweennessCentrality
7	0.45454545	0.29047619
5	0.51724138	0.42380952
4	0.48387097	0.4952381

Similar, but also different

Inconsistencies across centrality measures can be suggestive of interesting aspects of a node's position

	Low Degree	Low Closeness	Low Betweenness
High Degree		Embedded in cluster that is far from the rest of the network	Ego's connections are redundant - communication bypasses him/her
High Closeness	Key player tied to important important/active alters		Probably multiple paths in the network, ego is near many people, but so are many others
High Betweenness	Ego's few ties are crucial for network flow	Very rare cell. Would mean that ego monopolizes the ties from a small number of people to many others.	

Table source: James Moody

Summary

Centrality as a general term for measuring a node's position of "prominence" in the network

Degree: sheer connections

Closeness: shortest path distance

Betweenness: node's presence in shortest paths

Eigenvector: degree weighted by the degree of the neighbors

These measures are highly correlated, but conceptually distinct